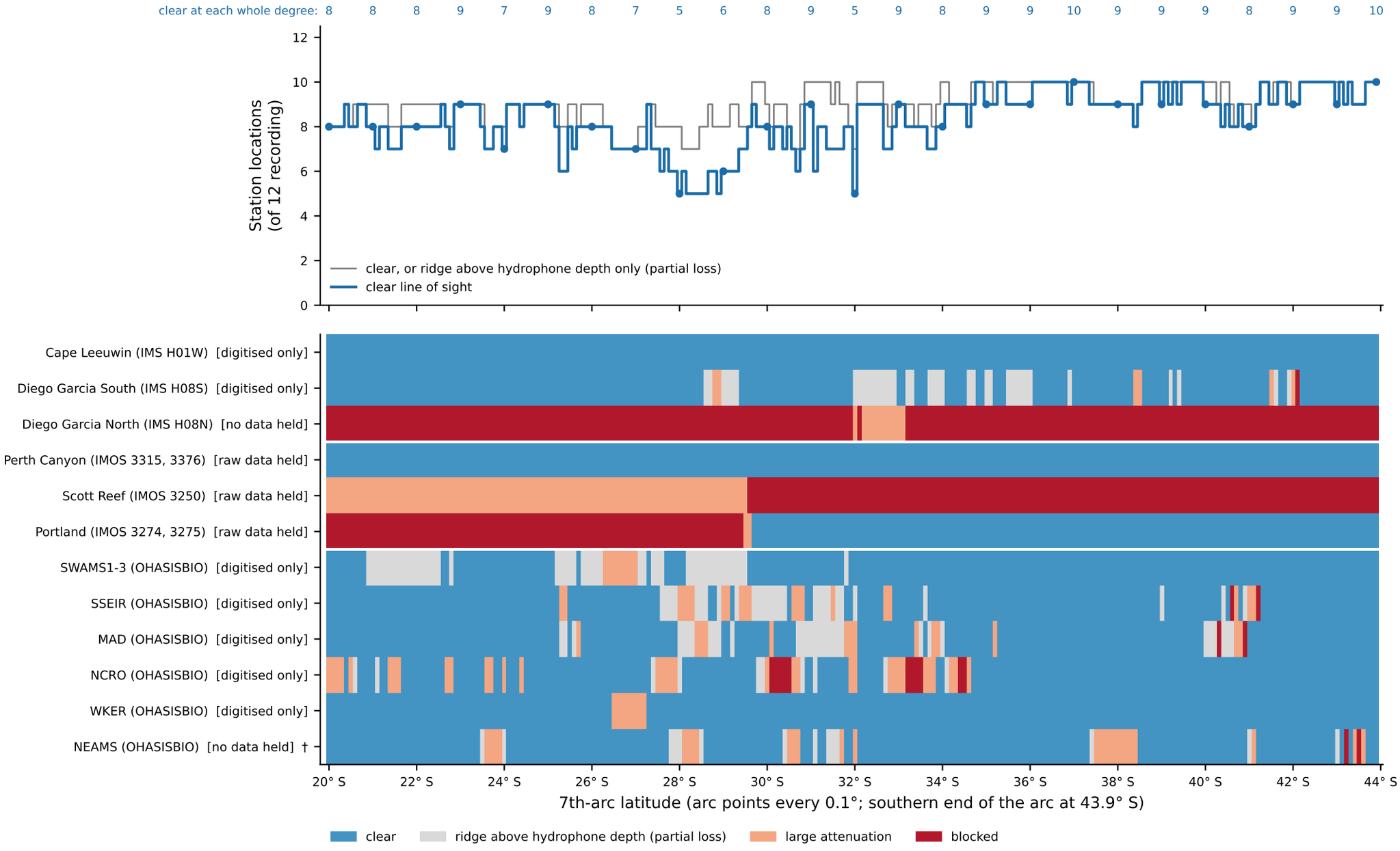
South of 33° S, 8–10 of the 12 recording station locations have clear line of sight at 98 % of 7th-arc points; north of 33° S, mid-ocean ridges cut this to 5–9



Plain-language note. For each point on the 7th arc, we follow the sea floor along the straight path to each station that was recording on 8 March 2014. 'Clear' means no land, no shallow water and no long ridge that rises into the deep sound channel. A station with clear line of sight can receive the sound, but it does not show that the sound was loud enough to detect; that is the next step. Stations at one place are counted once. † NEAMS was recording, but its data was lost when it was recovered.

Technical note. Source: hydro_data.sqlite tables arc_path and attribute (arc_line_of_sight.py, arc_station_count.py). 7th arc: small circle, centre 0.570° N 64.252° E, radius 44.501°. WGS84 geodesics on GEBCO 2026 (15 arc-s), ~1 km steps. Blocked: land or water shallower than min(receiver depth, 200 m) more than 20 km from the receiver. Large attenuation: >= 10 km shallower than 1,000 m more than 100 km from the receiver. Ridge above hydrophone depth: >= 10 km shallower than the hydrophone (deep receivers). Location class = most favourable instrument (Perth Canyon 3315/3376, Portland 3274/3275, SWAMS1-3). Not recording on 8 Mar 2014: HA04 Crozet (IMS) (installed Dec 2016, certified Jun 2017 (CTBTO)); IMOS Kangaroo Island, Tuncurry, Pilbara deployments (no 8 Mar 2014 files in the public IMOS archive (S3 listing, 23 Oct 2026)). Island seismometers that record T-phases are not included. Geometric screen, not transmission loss; IMOS loggers record 34 % of the time.